

# **K Balaram Senapati**

Bachelor of Technology – Computer Science Engineering

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## **Summary**

- Unity Developer with nearly 6 years of experience in Unity3D, Godot and Unreal Engine.
- Managed diverse technical aspects, including 2D and 3D game development, simulation development, AR, VR and MR, Block Chain game
- Engaged in requirement analysis, design and development of various products.
- Hands on experience in creating Cross platform Applications.

## **Key Responsibilities**

- Collaboration with teams to understand finalized requirements.
- Keeping the teams informed about progress regularly.
- Design and implementation of all end-to-end app functionalities.
- Troubleshooting and debugging technical issues with various debugging tools.
- Designing, writing and maintaining clean and robust code including code comments based on specifications.
- Enhance the app's quality and performance by gathering and analysing user reviews.
- Bug fixing, Designing App UI and Environment Enhancement.

## **Technical Skills**

Platform: Unity 3D, Godot, Unreal Engine, Arduino

Programing Languages: C#, Unreal Blueprints, GDScript, C, C++

Domain: 2D/3D, AR/VR/MR, Unity WebGL, Multiplayer, Block Chain, API REST & socket.io integration

Device: Oculus Quest, HTC Vive, Google Cardboard VR, Android Mobile

SDK: AR Foundation, Vuforia, Oculus, Steam VR, Photon, MRTK, VESPR Wallet, Eternal, Mirror Multiplayer, Logitech.

Version Controller: Git, Plastic SCM

## **Employment**

- Rams Creative Technologies Pvt Ltd: May 2025 - Present
- Innovana Thinklabs Ltd : Oct 2022 - Jun 2024
- Abhiwan Technology Pvt Ltd : Jun 2024 – Apr 2025
- Rams Creative Technologies Pvt Ltd: Aug 2021 – Jun 2022
- CHRP-India Pvt Ltd: Apr 2021 – Nov 2021
- Green Vision Solution Pvt Ltd: Aug 2019 - Apr 2021

## **Projects**

### **Vehicle for change (VFC)**

Developed an enterprise-scale Virtual Reality (VR) training and assessment ecosystem for automotive technical education, integrating immersive simulations with a web-based LMS/CMS for user tracking and analytics.

Designed and implemented interactive training modules including Oil Change, MLR, Brakes, and Steering & Suspension — each featuring 3D model-based procedures, voice guidance, and progress tracking.

Integrated Guide/Test modes with dynamic progress bars, checkpoints, and real-time performance analytics for skill evaluation.

Built interactive safety and maintenance simulations covering equipment handling, tool identification, and component replacements using Unity 3D and C#.

Developed authoring tools enabling administrators to upload and convert external 3D models (FBX/GLTF) into interactive, VR-ready content.

Contributed to a scalable, data-driven learning ecosystem bridging VR training, analytics, and content management.

**Reference Link:** <https://www.vehiclesforchange.org/>

### **Edu VR**

EDUVR is the first and only fully customized, curriculum-aligned education, assessment, and self-learning solution for schools. It makes education immersive, interactive, and engaging for students in Grades 3 to 10. Designed to work in School Mode (VR headsets) and Home Mode (cross-platform), it ensures seamless learning both inside and outside the classroom, covering key subjects like Mathematics, Science, English, Biology, Chemistry, and Physics.

It combine gamification, interactivity, immersion, and experiential learning to help student's master complex concepts with ease.

**Reference Link:** <https://eduvr-school.com/>

## BankOPoly

Developed a 2D block chain-integrated game blending casual gameplay mechanics with Web3 tokenization and NFT utilities. The project aimed to gamify DeFi and blockchain concepts — allowing players to earn, stake, and trade virtual assets securely on-chain while enjoying engaging 2D gameplay.

The game served as part of the Banker Labs ecosystem (“The Bank of Cardano”), combining financial education, token rewards, and interactive player progression through block chain-backed game logic.

**Reference Link:** <https://www.bankerlabs.io/>

## Levitate

Levitate is a virtual reality application designed to deliver immersive guided meditation and mindfulness experiences. The platform blends positive psychology, traditional mindfulness techniques, and spatial computing to help users reduce stress, improve focus, enhance well-being, and support better sleep.

The app transports the user into serene virtual environments (e.g. natural landscapes, contemplative spaces) where guided breathing, visualization, and relaxation exercises are performed in VR.

**Reference Link:** <https://www.levitate.world/>

## Your Tarot Life

“Your Tarot Life” is an immersive XR / Meta Quest experience that blends augmented storytelling, interactive tarot reading, and virtual metaphysical environments. It allows users to navigate mystical worlds, engage with tarot cards, reveal story paths, and experience narrative-driven interactions in a VR/AR setting via the Meta platform.

**Reference Link:** <https://www.meta.com/en-gb/experiences/your-tarot-life/7039326086112973/>

## 3D Snake vs Block Beats

Developed a modern, arcade-style snake / slither mobile game for Android that reimagines the classic “Snake” gameplay with features like dynamic growth, competitive mechanics, and user engagement systems. The game combines addictive arcade controls with monetization, leader boards, and performance analytics.

**Reference Link:** <https://play.google.com/store/apps/details?id=ig.snake.blocks.slither.free.games.android>

## **The Jumpers - Mega Jump Game**

“The Jumpers – Mega Jump Game” is a casual, arcade-style mobile game released on Android with over 10K installs.

Players tap to make a hero character jump through tile platforms, dodge obstacles (chainsaws, bombs, etc.), collect coins, and reach exit doors.

The game features rich visual effects, challenging level designs, and engaging mechanics to encourage replay and high score chasing.

### **Reference Link:**

<https://play.google.com/store/apps/details?id=com.jumpers.casual.arcade.adventure.android>

## **Atyaf – VR Aviation**

Atyaf is a next-generation Virtual Reality (VR) training platform for the aviation industry, developed to revolutionize the way aircraft maintenance personnel, engineers, and aviation students learn critical operations.

The system simulates real-world maintenance environments, enabling users to train, practice, and perform aircraft procedures in an immersive, risk-free setting.

The simulation includes interactive training modules such as:

- Aircraft Marshalling
- Air Start Unit (ASU) Operation
- Air Conditioning Unit (ACU) Handling
- Passenger Step and Ground Support Equipment Operations
- Pushback Tractor

By replicating real aircraft scenarios, Atyaf enhances procedural accuracy, safety awareness, and technical proficiency while reducing training costs and equipment downtime.

## **MetaBLOQS**

MetaBLOQS is a decentralized metaverse platform built on the XinFin XDC Network, designed to provide users with immersive experiences for networking, learning, and entertainment. The platform offers a virtual world where users can create, own, and monetize their experiences using the native BLOQS token. MetaBLOQS aims to democratize education, promote authentic connections, and enable users to participate in a decentralized economy of trust.

**Reference Link:** <https://metablogs.com/>

## **Sify – Welding Simulation**

Developed a Virtual Reality (VR) welding simulation for Sify to train technicians and students in welding operations in a risk-free, immersive environment. The simulator replicates real-world welding scenarios, enabling users to practice procedural accuracy, hand-eye coordination, and safety compliance without the hazards of actual welding.

The system is designed to enhance skill acquisition, reduce training costs, and provide real-time performance analytics for trainees and instructors.

## **TATA Power BlackStart – Rout 1 & Rout2**

Developed a Virtual Reality (VR) training system for TATA Power to simulate BlackStart procedures in power plants. The VR platform provides a safe, immersive environment for operators and engineers to practice complex startup sequences of power generation units without risking equipment or personnel safety.

## **NRL VR Training Simulation**

Developed a Virtual Reality (VR) training simulation for Numaligarh Refinery Limited (NRL), aiming to enhance the safety and efficiency of refinery operations through immersive training experiences. The simulation replicates critical refinery processes, allowing operators and maintenance personnel to practice procedures in a controlled, risk-free environment.

The simulation includes interactive training modules such as:

- Major Spillage Protection
- Minor Spillage Protection
- Loading Oil Lubricant
- Pump Operation

**Reference Link:** <https://www.nrl.co.in/>

## **DSM – Plant Tour**

Developed a Virtual Reality (VR) plant tour application for DSM to provide employees, trainees, and visitors with an immersive and interactive experience of the manufacturing facility.

The application allows users to explore complex industrial processes, machinery, and workflows in a risk-free virtual environment, enhancing familiarity, safety awareness, and operational understanding.

## **VR – OP**

Developed a Virtual Reality (VR) and IoT-integrated Observation Post (OP) training simulation for the Indian Army.

The system enables soldiers to observe border areas, detect targets, and practice surveillance protocols in a safe, immersive, and controlled virtual environment, reducing risk and improving operational readiness.

The simulation includes interactive training modules such as:

- VR Simulation Development
- IoT & Sensor Integration
- Training & Assessment Modules
- Immersion & Optimization
- User Experience & Safety

## **Lorros (Long-Range Reconnaissance and Observation System)**

Developed a training simulation for LORROS, designed to train military personnel in long-range reconnaissance, surveillance, and observation operations.

The system provides an immersive, risk-free environment for personnel to practice operating high-precision observation equipment, analyzing data, and making tactical decisions in simulated border or battlefield scenarios.

The simulation includes interactive training modules such as:

- Environment & Simulation Development
  - Created realistic 3D terrains, observation posts, and target zones for long-range surveillance training.
- IoT & Sensor Integration
  - Integrated real-time sensor data and simulated telemetry to emulate live LORROS operations.
  - Developed systems to track moving targets and provide feedback on observation accuracy.
- Training & Assessment Modules
  - Designed guided training modules covering reconnaissance protocols, target detection, and reporting procedures.
  - Implemented real-time analytics to evaluate trainee performance, including reaction time, observation accuracy, and decision-making.
- Immersion & Realism
  - Added spatial audio, weather effects, and realistic lighting to replicate operational conditions.
  - Optimized performance for smooth and low-latency interactions on standalone headsets.

- User Experience & Safety
  - Developed intuitive hand interactions and device controls to simulate real observation instruments.
  - Created scenarios to train personnel on high-risk reconnaissance operations safely in Windows Platform

## New Generation

Developed a Windows-based driving simulation platform with 3 Degrees of Freedom (3DOF) and 6 Degrees of Freedom (6DOF) motion platforms for the Indian Army.

The system enables trainees to practice driving manoeuvres and tactical operations in a fully immersive, hardware-integrated environment, replicating realistic vehicle dynamics, terrain conditions, and operational scenarios without risk.

The simulation includes interactive training modules such as:

- Simulation Development & Hardware Integration
  - Developed driving simulation software with realistic vehicle dynamics and motion feedback.
  - Integrated 3DOF and 6DOF motion platforms for synchronized physical motion corresponding to virtual driving scenarios.
- Interactive Driving Scenarios
  - Created multiple training scenarios including off-road driving, convoy manoeuvres, and obstacle navigation.
  - Implemented real-time vehicle physics, terrain modelling, and environmental conditions.
- Sensor & Control Systems Integration
  - Connected hardware controls (steering, pedals, gear systems) with the simulation engine.
  - Integrated sensors for motion feedback, telemetry, and performance monitoring.
- Performance Monitoring & Assessment
  - Developed analytics to evaluate trainee performance, including speed control, manoeuvre accuracy, and response time.
  - Enabled instructors to monitor real-time progress and generate reports.
- Optimization & Realism
  - Ensured high-fidelity simulation with smooth motion transitions, low latency, and realistic feedback.
  - Applied visual effects, lighting, and spatial audio for immersive experience.

## Carrier Mortar Simulation Platform

Developed a Windows-based Carrier Mortar simulator with integrated hardware systems for the Indian Army.

The system provides immersive, risk-free training for operators to practice firing sequences, aiming procedures, and tactical operations, while replicating the real behavior of carrier-mounted mortar systems.

The simulation includes interactive training modules such as:

- **Simulation Development & Hardware Integration**
  - Developed a high-fidelity simulation engine replicating the dynamics of carrier mortar operation.
  - Integrated hardware interfaces for controls, sensors, and motion feedback, providing realistic operational response.
- **Interactive Training Modules**
  - Implemented modules for aiming, targeting, firing, and reload sequences.
  - Simulated battlefield conditions, including terrain variation, environmental effects, and moving targets.
- **Sensor & Feedback Systems**
  - Integrated real-time telemetry and motion feedback from hardware components.
  - Enabled haptic and visual feedback to replicate live firing experience safely.
- **Performance Monitoring & Analytics**
  - Developed real-time tracking of trainee performance including accuracy, response time, and procedural compliance.
  - Provided analytics dashboards for instructors to evaluate and improve training effectiveness.
- **Optimization & Realism**
  - Applied realistic visuals, spatial audio, and optimized motion feedback for immersive experience.
  - Ensured smooth, low-latency integration between software simulation and hardware motion systems.